

Non-Parametric Statistics

C5025HF

Dr K.J. Mhango

Welcome

What You Will Learn Today

- Parametric vs non-parametric intuition
- Ranking logic and distribution-free ideas
- Mann–Whitney, Wilcoxon, Kruskal–Wallis
- Correlations: Spearman & Kendall
- Exact tests (Fisher)

1: Why Non-Parametrics?

When parametric assumptions break

- Non-normal data
- Skewed distributions
- Ordinal measurements
- Heterogeneous variances
- Small samples
- Outliers

Ranking-based thinking: The “Race” Analogy

Instead of looking at exact times (values), look at finishing positions (ranks).

The Intuition: Team A vs Team B (Independent)

Imagine a race between two teams.

- **Parametric (t-test):** Compares the **Average Time** of Team A vs Team B.
 - If **one** runner in Team A takes 5 hours (outlier), Team A's *average* is ruined.
- **Non-Parametric (Mann-Whitney):** Compares the **Finishing Positions** (Ranks).
 - **It combines everyone** (both teams) into one single lineup from 1st to last.
 - Then it looks at where Team A falls vs Team B.
 - If that slow runner takes 5 hours or 5 days, they are still just **“Last Place”**. The rank (last) is the same.

The Intuition: Before vs After (Paired)

Imagine the **same runner** running twice (Race 1 vs Race 2).

- **Parametric (Paired t-test):** Calculates the improvement (Time 1 - Time 2) for each runner, then averages them.
 - One runner improving by 5 hours makes the *whole group* look vastly improved on average.
- **Non-Parametric (Wilcoxon Signed-Rank):**
 - It calculates the change for each runner.
 - **It ranks these changes by size** (ignoring direction). Small change = Rank 1, Huge change = Rank 100.
 - Then it asks: **“Are the biggest ranks associated with getting faster or getting slower?”**
 - If one runner improves by 5 hours, they get the highest rank, but they don't pull the *magnitude* of the sum to infinity. It caps the influence of the outlier.

Why this is useful

- **Robust to Outliers:** Extreme values don't pull the results (unlike the mean).
- **Skew doesn't matter:** We are testing the order, not the bell curve.
- **Any units work:** cm, mm, log-transformed, or “Likert scales” — as long as you can say “A > B”, you can rank.

What the tests actually ask

- **Mann–Whitney (2 groups):** If I pick one random value from Group A and one from Group B, is A likely to be higher than B?
- **Wilcoxon (Paired):** It looks at the *change* in each pair. Do the **big changes** tend to be increases or decreases?
 - (It ranks the size of the changes, then checks if the positive ranks outweigh the negative ranks).
- **Kruskal–Wallis (>2 groups):** Do some groups systematically rank higher than others?

What to report

- **Descriptive stats:** Medians and Interquartile Ranges (IQR) for each group.
- **Effect size / Direction:** Explain the *direction* of the difference in plain English.
 - Instead of just “ $p < 0.05$ ”, say: “Species A tends to have higher values than Species B.”
 - Optional but good: “The probability that a random value from Group A exceeds Group B is approx X%.”
- **Visuals:** Always include a boxplot or jitter plot to show the spread and overlap.

2: Choosing the Right Test

Cheat Sheet

Design	Parametric	Non-parametric
2 independent groups	t-test	Mann–Whitney
2 paired measures	paired t	Wilcoxon signed-rank
>2 independent groups	ANOVA	Kruskal–Wallis
Correlation	Pearson	Spearman, Kendall
2×2 association	² test	Fisher exact

3: t-test Using *iris*

Goal

Compare **Sepal.Length** between two species. We subset to two species for a t-test:

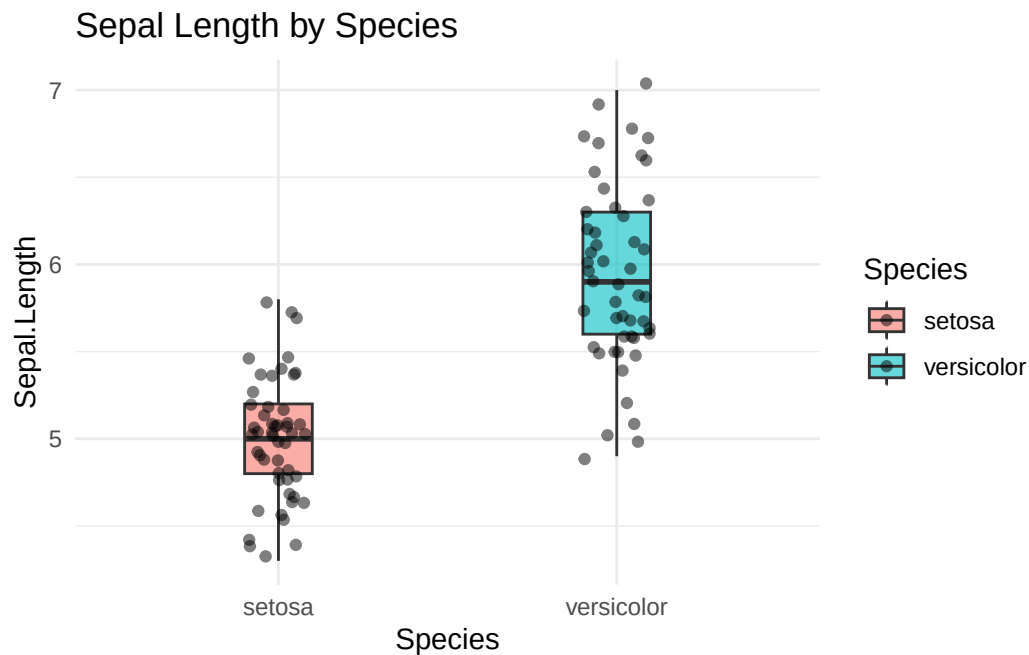
```
iris2 <- subset(iris, Species %in% c("setosa", "versicolor"))
```

Visualise

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(iris2, aes(x = Species, y = Sepal.Length, fill = Species)) +  
  geom_boxplot(alpha = 0.6, width = 0.2) +  
  geom_jitter(width = 0.1, alpha = 0.5) +  
  theme_minimal() +  
  ggtitle("Sepal Length by Species")
```



Parametric t-test Assumptions

1. Independence

- Observations must be independent.
- In *iris*, each flower is a unique specimen → OK.

2. Normality of the outcome within each group

- Test with Shapiro–Wilk.

```
shapiro.test(iris2$Sepal.Length[iris2$Species=="setosa"])
```

Shapiro-Wilk normality test

```
data:  iris2$Sepal.Length[iris2$Species == "setosa"]  
W = 0.9777, p-value = 0.4595
```

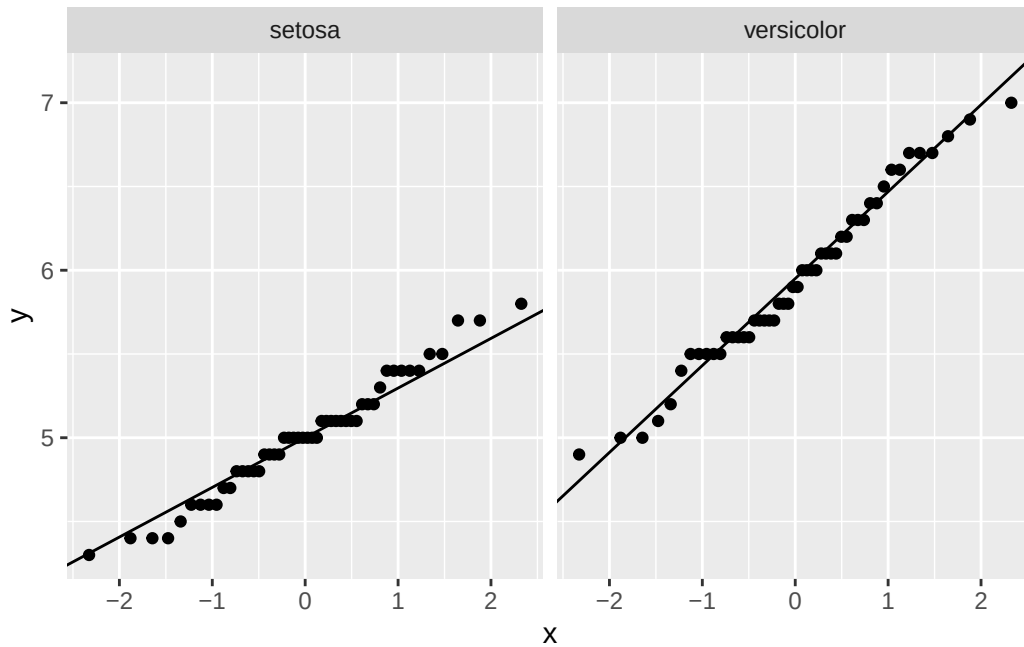
```
shapiro.test(iris2$Sepal.Length[iris2$Species=="versicolor"])
```

Shapiro-Wilk normality test

```
data:  iris2$Sepal.Length[iris2$Species == "versicolor"]  
W = 0.97784, p-value = 0.4647
```

- If $p < 0.05$ in either group, normality is questionable for that group.
 - Also use Q–Q plot:

```
library(ggplot2)  
ggplot(iris2, aes(sample = Sepal.Length)) + stat_qq() + stat_qq_line() + facet_wrap(~Species)
```



- Points close to the straight line suggest normality; strong bends or S-shapes suggest non-normality.

3. Homogeneity of variances (Levene test)

```
library(car)
```

Loading required package: carData

```
leveneTest(Sepal.Length ~ Species, data = iris2)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group 1	8.1727	0.005196	**
	98		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- If $p < 0.05$, variances differ notably between the two species.

What if assumptions fail?

- Non-normality → use **Mann–Whitney (Wilcoxon rank-sum)**.
- Heterogeneous variances → Welch t-test OR non-parametric.

Parametric test

```
t.test(Sepal.Length ~ Species, data = iris2)
```

Welch Two Sample t-test

data: Sepal.Length by Species

t = -10.521, df = 86.538, p-value < 2.2e-16

alternative hypothesis: true difference in means between group setosa and group versicolor is not equal to 0

95 percent confidence interval:

-1.1057074 -0.7542926

sample estimates:

mean in group setosa	mean in group versicolor
5.006	5.936

- Look for the p-value and the confidence interval; with normality and similar variances, this is appropriate.

Non-parametric alternative (if assumptions break)

```
wilcox.test(Sepal.Length ~ Species, data = iris2)
```

Wilcoxon rank sum test with continuity correction

data: Sepal.Length by Species

W = 168.5, p-value = 8.346e-14

alternative hypothesis: true location shift is not equal to 0

- Use this when data are skewed or have outliers; the p-value tests if one group tends to have higher values.

Example output and interpretation (R)

```
# Run both to compare

tt_res <- t.test(Sepal.Length ~ Species, data = iris2)
mw_res <- wilcox.test(Sepal.Length ~ Species, data = iris2, exact = FALSE)

tt_res
```

Welch Two Sample t-test

```
data: Sepal.Length by Species
t = -10.521, df = 86.538, p-value < 2.2e-16
alternative hypothesis: true difference in means between group setosa and group versicolor is not equal to 0
95 percent confidence interval:
 -1.1057074 -0.7542926
sample estimates:
 mean in group setosa mean in group versicolor
                5.006                5.936
```

```
mw_res
```

Wilcoxon rank sum test with continuity correction

```
data: Sepal.Length by Species
W = 168.5, p-value = 8.346e-14
alternative hypothesis: true location shift is not equal to 0
```

```
# Print key summaries for the slide
cat("t-test p-value:", format.pval(tt_res$p.value, digits = 3), "\n")
```

```
t-test p-value: <2e-16
```

```
cat("Mann-Whitney p-value:", format.pval(mw_res$p.value, digits = 3), "\n")
```

```
Mann-Whitney p-value: 8.35e-14
```

- Interpret the p-values and medians.
- Interpretation: If data look non-normal or have outliers, prefer Mann-Whitney. Report simply, e.g., “Species A tends to have longer sepals than Species B.”

Paired t-test Using *iris*

Goal

Compare **Sepal.Length** vs **Sepal.Width** for the same flowers. Since these measurements come from the *same* flower, they are paired.

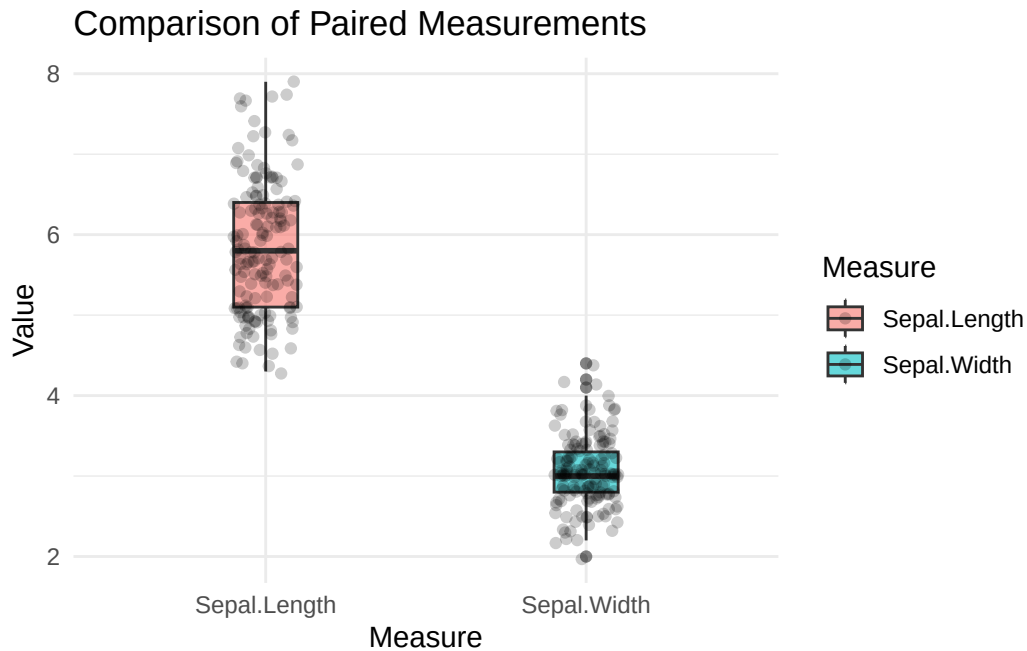
Visualise

```
# Reshape data to long format for plotting
library(tidyr)
```

Warning: package 'tidyr' was built under R version 4.4.3

```
iris_long <- pivot_longer(iris,
                           cols = c("Sepal.Length", "Sepal.Width"),
                           names_to = "Measure",
                           values_to = "Value")

ggplot(iris_long, aes(x = Measure, y = Value, fill = Measure)) +
  geom_boxplot(alpha = 0.6, width = 0.2) +
  geom_jitter(width = 0.1, alpha = 0.2) +
  theme_minimal() +
  ggtitle("Comparison of Paired Measurements")
```



Paired t-test Assumptions

1. Independence of pairs

- Each flower is independent of others → OK.

2. Normality of the differences

- We care about the distribution of $(Sepal.Length - Sepal.Width)$.

```
diffs <- iris$Sepal.Length - iris$Sepal.Width
shapiro.test(diffs)
```

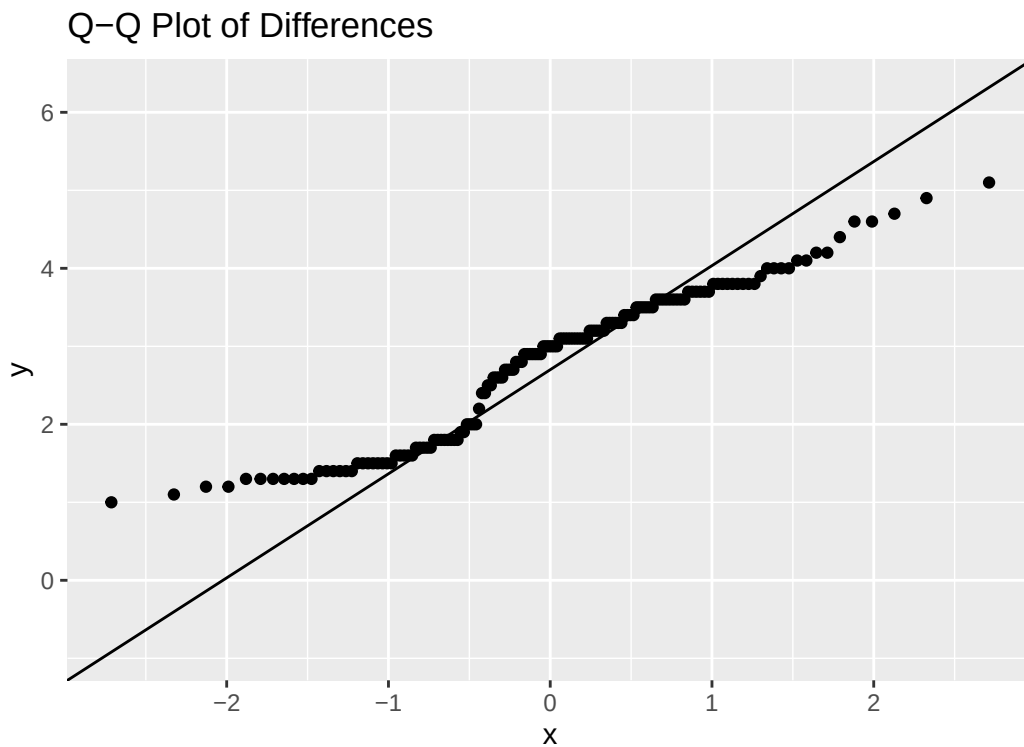
Shapiro-Wilk normality test

data: diffs

W = 0.94628, p-value = 1.628e-05

- If $p < 0.05$, the differences are not normally distributed.
- Check visually:

```
ggplot(data.frame(diffs), aes(sample = diffs)) + stat_qq() + stat_qq_line() +
  ggtitle("Q-Q Plot of Differences")
```



Parametric paired test

```
t.test(iris$Sepal.Length, iris$Sepal.Width, paired = TRUE)
```

Paired t-test

```
data: iris$Sepal.Length and iris$Sepal.Width
t = 34.815, df = 149, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 2.627874 2.944126
sample estimates:
mean difference
      2.786
```

- Checks if the mean difference is non-zero.

Non-parametric alternative (if assumptions break)

Wilcoxon Signed-Rank Test

```
wilcox.test(iris$Sepal.Length, iris$Sepal.Width, paired = TRUE)
```

Wilcoxon signed rank test with continuity correction

data: iris\$Sepal.Length and iris\$Sepal.Width

V = 11325, p-value < 2.2e-16

alternative hypothesis: true location shift is not equal to 0

- Tests if the median difference is non-zero (symmetric assumption) or if the distribution of differences is symmetric around zero.

Example output and interpretation (R)

```
pair_t <- t.test(iris$Sepal.Length, iris$Sepal.Width, paired = TRUE)
pair_w <- wilcox.test(iris$Sepal.Length, iris$Sepal.Width, paired = TRUE)
```

```
pair_t
```

Paired t-test

data: iris\$Sepal.Length and iris\$Sepal.Width

t = 34.815, df = 149, p-value < 2.2e-16

alternative hypothesis: true mean difference is not equal to 0

95 percent confidence interval:

2.627874 2.944126

sample estimates:

mean difference

2.786

```
pair_w
```

Wilcoxon signed rank test with continuity correction

```
data: iris$Sepal.Length and iris$Sepal.Width
```

```
V = 11325, p-value < 2.2e-16
```

```
alternative hypothesis: true location shift is not equal to 0
```

- Interpretation: “Sepal length is significantly different from sepal width within the same flowers.”

Chi-square Test Using *iris*

Goal

Test independence between **Species** and a discretised version of **Sepal.Width**. We bin Sepal.Width:

```
iris$WidthClass <- cut(iris$Sepal.Width, breaks = 3, labels = c("Narrow", "Medium", "Wide"))
tab <- table(iris$Species, iris$WidthClass)
```

Visualise

```
ggplot(iris, aes(x = Species, fill = WidthClass)) +
  geom_bar(position = "fill") +
  theme_minimal() +
  labs(y = "Proportion", title = "Width Class Distribution by Species") +
  scale_fill_brewer(palette = "Pastel1")
```



Chi-square Assumptions

1. Independence of observations → OK.
2. Expected cell counts ≥ 5 in most cells.

```
chisq.test(tab)$expected
```

	Narrow	Medium	Wide
setosa	15.66667	29.33333	5
versicolor	15.66667	29.33333	5
virginica	15.66667	29.33333	5

If expected counts are too small → chi-square invalid. - Scan the table: if any expected cell count is < 5, prefer Fisher's Exact Test.

Parametric-style categorical test

```
chisq.test(tab)
```

Pearson's Chi-squared test

data: tab

X-squared = 45.125, df = 4, p-value = 3.746e-09

- Chi-square p-value tests association; warnings about small counts mean the test may be unreliable.

Non-parametric alternative when assumptions fail

Fisher's Exact Test

Works even with small expected cell counts.

```
fisher.test(tab)
```

Fisher's Exact Test for Count Data

data: tab

p-value = 8.429e-11

alternative hypothesis: two.sided

- Fisher's p-value is reliable even with small expected counts.

Example output and interpretation (R)

```
chisq_res <- suppressWarnings(chisq.test(tab)) # warning if small counts
```

```
fisher_res <- fisher.test(tab)
```

```
chisq_res
```

Pearson's Chi-squared test

data: tab

X-squared = 45.125, df = 4, p-value = 3.746e-09

```
fisher_res
```

Fisher's Exact Test for Count Data

```
data: tab  
p-value = 8.429e-11  
alternative hypothesis: two.sided
```

```
chisq_expected <- chisq_res$expected  
  
chisq_expected
```

	Narrow	Medium	Wide
setosa	15.66667	29.33333	5
versicolor	15.66667	29.33333	5
virginica	15.66667	29.33333	5

- Check expected counts; if any are < 5 , prefer Fisher's result.
- Interpretation: "Species and width class show evidence of association."

ANOVA Using *iris*

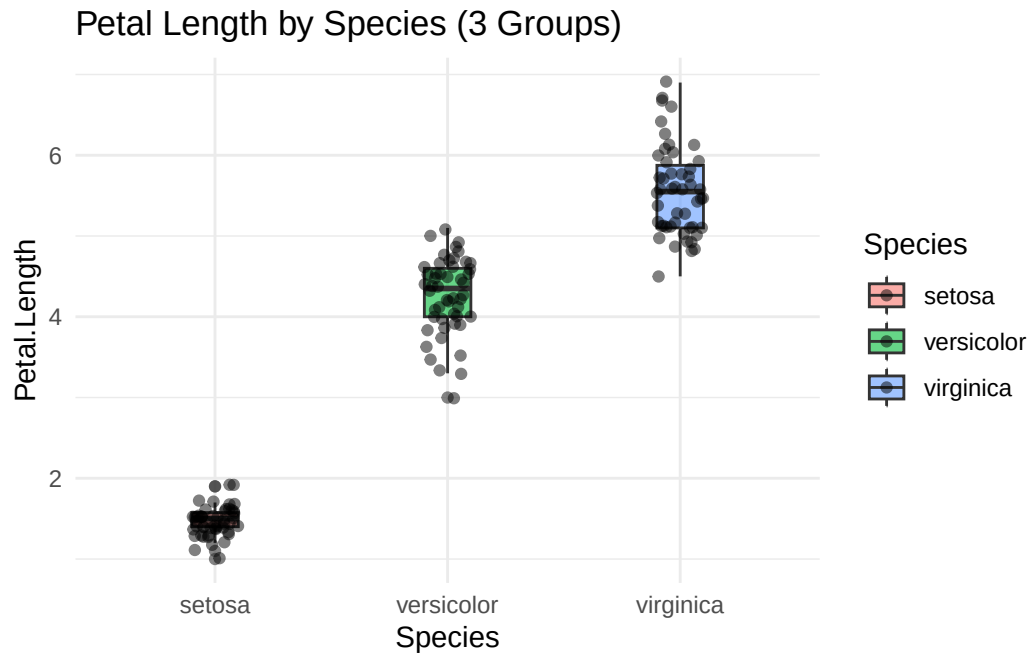
Goal

Compare **Petal.Length** across all three species.

```
iris$Species <- factor(iris$Species)
```

Visualise

```
ggplot(iris, aes(x = Species, y = Petal.Length, fill = Species)) +  
  geom_boxplot(alpha = 0.6, width=0.2) +  
  geom_jitter(width = 0.1, alpha = 0.5) +  
  theme_minimal() +  
  ggtitle("Petal Length by Species (3 Groups)")
```

ANOVA Assumptions

1. Independence → by design.
2. Normality within each group

```
by(iris$Petal.Length, iris$Species, shapiro.test)
```

```
iris$Species: setosa
```

```
Shapiro-Wilk normality test
```

```
data: dd[x, ]
```

```
W = 0.95498, p-value = 0.05481
```

```
-----  
iris$Species: versicolor
```

```
Shapiro-Wilk normality test
```

```
data: dd[x, ]
```

```
W = 0.966, p-value = 0.1585
```

```
-----  
iris$Species: virginica
```

Shapiro-Wilk normality test

```
data: dd[x, ]
```

```
W = 0.96219, p-value = 0.1098
```

- If some species have $p < 0.05$, normality is doubtful for those groups.

3. Homogeneity of variances (Levene)

```
leveneTest(Petal.Length ~ Species, data = iris)
```

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	2	19.48	3.129e-08 ***
	147		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- If $p < 0.05$, variances differ across species.

If either normality or variance equality fails → cannot trust ANOVA F-test.

Parametric ANOVA

```
summary(aov(Petal.Length ~ Species, data = iris))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Species	2	437.1	218.55	1180	<2e-16 ***
Residuals	147	27.2	0.19		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- In the ANOVA table, check the row for Species: $\text{Pr}(>F)$ is the p-value for group differences.

Non-parametric alternative

Kruskal–Wallis Test

```
kruskal.test(Petal.Length ~ Species, data = iris)
```

Kruskal-Wallis rank sum test

data: Petal.Length by Species

Kruskal-Wallis chi-squared = 130.41, df = 2, p-value < 2.2e-16

- Kruskal–Wallis p-value tests whether at least one group tends to have higher ranks than the others.

If significant → perform post-hoc Dunn tests.

```
if (!requireNamespace("FSA", quietly = TRUE)) {  
  cat("Package 'FSA' not installed; skipping Dunn post-hoc tests. Install with install.packages('FSA')")  
} else {  
  library(FSA)  
  dunnTest(Petal.Length ~ Species, data = iris, method = "bonferroni")  
}
```

Registered S3 methods overwritten by 'FSA':

method	from
confint.boot	car
hist.boot	car

Warning: package 'FSA' was built under R version 4.4.3

FSA v0.10.0. See citation('FSA') if used in publication.

Run fishR() for related website and fishR('IFAR') for related book.

Attaching package: 'FSA'

The following object is masked from 'package:car':

bootCase

Dunn (1964) Kruskal-Wallis multiple comparison

p-values adjusted with the Bonferroni method.

	Comparison	Z	P.unadj	P.adj
1	setosa - versicolor	-5.862997	4.545875e-09	1.363763e-08
2	setosa - virginica	-11.418385	3.384664e-30	1.015399e-29
3	versicolor - virginica	-5.555388	2.769957e-08	8.309872e-08

- Post-hoc Dunn tests (when available) show which pairs differ; use adjusted p-values to report significant pairs. If skipped, you can still report the Kruskal-Wallis result and show group medians/IQRs.

Example output and interpretation (R)

```
aov_res <- aov(Petal.Length ~ Species, data = iris)
kw_res <- kruskal.test(Petal.Length ~ Species, data = iris)

summary(aov_res)
```

```
          Df Sum Sq Mean Sq F value Pr(>F)
Species    2  437.1   218.55    1180 <2e-16 ***
Residuals 147   27.2    0.19
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
kw_res
```

Kruskal-Wallis rank sum test

data: Petal.Length by Species

Kruskal-Wallis chi-squared = 130.41, df = 2, p-value < 2.2e-16

```
# Extract and print concise p-values for the slide
s <- summary(aov_res)[[1]]
aov_p <- s[["Pr(>F)"]][1]
```

- Interpretation: With skew/outliers or non-normality, rely on Kruskal-Wallis. If significant, follow with pairwise comparisons (e.g., Dunn tests) and report which species differ.

Correlation Using *iris*

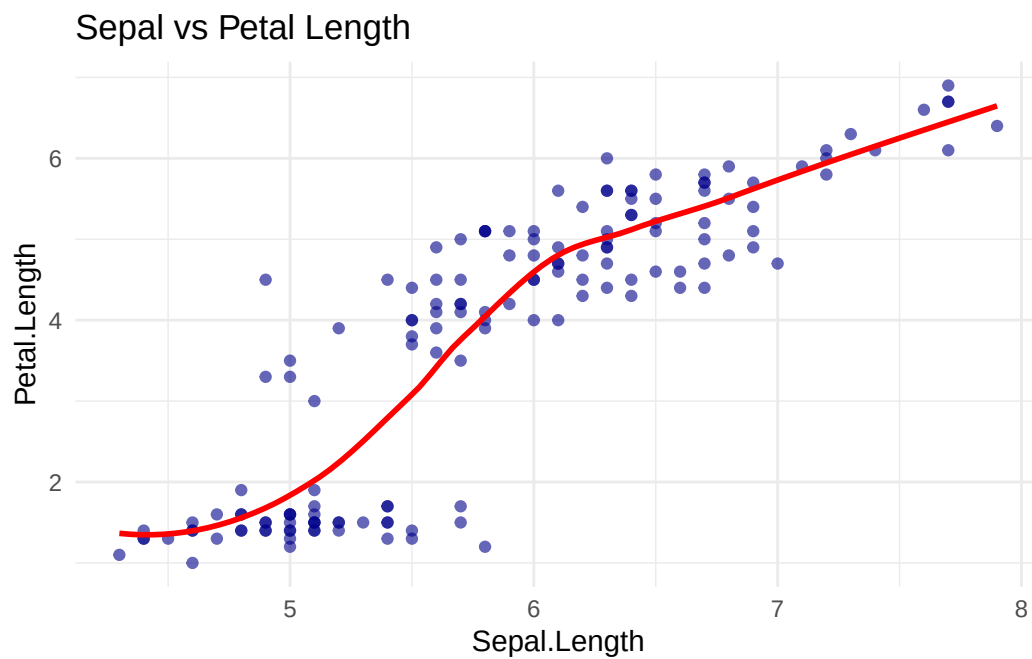
Goal

Assess the relationship between **Sepal.Length** and **Petal.Length**.

Visualise

```
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length)) +  
  geom_point(alpha = 0.6, color = "darkblue") +  
  geom_smooth(method = "loess", color = "red", se = FALSE) +  
  theme_minimal() +  
  ggtitle("Sepal vs Petal Length")
```

`geom\_smooth()` using formula = 'y ~ x'



Parametric Correlation (Pearson) Assumptions

1. Linearity

- The relationship should be a straight line.

2. Normality

- Both variables should be normally distributed.

```
shapiro.test(iris$Sepal.Length)
```

Shapiro-Wilk normality test

```
data:  iris$Sepal.Length  
W = 0.97609, p-value = 0.01018
```

```
shapiro.test(iris$Petal.Length)
```

Shapiro-Wilk normality test

```
data:  iris$Petal.Length  
W = 0.87627, p-value = 7.412e-10
```

- If $p < 0.05$, normality assumption is violated.

Parametric test (Pearson)

```
cor.test(iris$Sepal.Length, iris$Petal.Length, method = "pearson")
```

Pearson's product-moment correlation

```
data:  iris$Sepal.Length and iris$Petal.Length  
t = 21.646, df = 148, p-value < 2.2e-16  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
0.8270363 0.9055080
```

sample estimates:

cor

0.8717538

- Tests for linear correlation.

Non-parametric alternative (if assumptions break)

Spearman's Rank Correlation

```
cor.test(iris$Sepal.Length, iris$Petal.Length, method = "spearman")
```

Warning in cor.test.default(iris\$Sepal.Length, iris\$Petal.Length, method = "spearman"): Cannot compute exact p-value with ties

Spearman's rank correlation rho

data: iris\$Sepal.Length and iris\$Petal.Length

S = 66429, p-value < 2.2e-16

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

0.8818981

- Uses ranks; tests for *monotonic* relationship (doesn't have to be a straight line, just consistently increasing or decreasing).

Kendall's Tau

```
cor.test(iris$Sepal.Length, iris$Petal.Length, method = "kendall")
```

Kendall's rank correlation tau

data: iris\$Sepal.Length and iris\$Petal.Length

z = 12.647, p-value < 2.2e-16

alternative hypothesis: true tau is not equal to 0

sample estimates:

tau

0.7185159

- Better for small samples or many ties.
- Interpretation: “There is a strong positive monotonic correlation between sepal length and petal length.”

Summary: Use a non-parametric test when:

- Data are **clearly non-normal** (Shapiro–Wilk fails; Q–Q plots bend strongly).
- **Variances differ strongly** across groups (Levene fails).
- Data are **ordinal** (ranks, Likert scores).
- There are **outliers** that distort means.
- Sample size is **small**, making CLT unreliable.
- Group distributions have **different shapes**, not just shifted means.